The pxMEM package

Chapters, annexes and selective reuse in P3CTeX

Scvria
P3CTeX / UOC PEC Repository

2026/10/22 – Version 0.2

Abstract

pxMEM adds report-like chapter headings, annexes, and selective reuse of material from existing LaTeX sources. It is generic: a memory/report does not need to be a TFG. Source documents remain independent deliverables.

Contents

1 Activation and scope

Under P3CTeX, select MEM in the class options, before the class loads its feature modules:

```
\documentclass[MEM,CAT,toc]{P3CTeX}   % generic memory with indexes
\documentclass[TFG,MEM,PR,CAT]{P3CTeX} % TFG memory with cover and indexes
```

MEM does not select TFG; neither option implies the other. MEM is a load-time profile. Do not attempt to activate it later through \PECTeXconfig{MEM}. In non-P3CTeX classes, the module can be loaded with \usepackage{pxMEM}; its coloured chapter heading assumes the primaryColor colour supplied by P3CTeX/pxGDX.

P3CTeX continues to derive from article. Ordinary PR, PAC, PEC and CA documents retain their existing section and cover behaviour; MEM provides chapters only when selected.

2 Memory preliminary structure

When the class TOC is enabled, pxMEM automatically follows it with the lists of figures and tables. It does not load or depend on pxTFG; a generic memory never displays TFG-specific material. The MEM option alone does not enable the class cover or TOC; use PR or toc where those are wanted.

```
\documentclass[MEM,CAT,toc]{P3CTeX}
\begin{document}
\chapter{Introduction}
\end{document}
```

TFG-specific material is inserted by `pxTFG` only when TFG is selected, independently of generic MEM indexes:

```
\documentclass[TFG,MEM,PR,CAT]{P3CTeX}
\begin{document}
\chapter{Introduccio}
\end{document}
```

For older or custom documents with automatic TOC disabled, the optional `\pxMEMpreliminaries{<content>}` command prints caller-supplied preliminary content and the class TOC with MEM indexes. Already printed indexes are not printed again. Ordinary documents do not need this call. Chapters, glossary entries and bibliography remain under the author's control. The complete UOC TFG example is in `examples/UOCTemplates/TFGTemplate/MEM/MEM.tex`.

3 Chapters and annexes

3.1 `\chapter{title}`

Introduces a numbered, page-breaking chapter. The chapter appears in the table of contents; sections are numbered within the chapter (for example, `2.1`). This is a minimal article-based chapter implementation, not a complete `report` or `book` class.

```
\chapter{Introduccio}
\section{Objectius}
```

3.2 `\pxMEMannex{title}{content}`

Starts a titled annex chapter and typesets its content. On the first call it switches to appendix chapter numbering (A, B, ...); subsequent calls produce B, C, ... After this switch, later `\chapter` calls also use appendix numbering. Place annexes after the main chapters.

```
\pxMEMannex{Technical evidence}{
  \pxMEMinclude{PR2}{evidence}
}
```

The regular `\appendix` command also switches to lettered chapter numbers, but `\pxMEMannex` normally handles that transition itself.

4 Selective content reuse

A PR/PAC/PEC/CA can mark potentially reusable material without importing `pxMEM` or knowing which future report might use it.

4.1 1. Mark a region in the source

Use exact comment markers on their own lines, surrounding ordinary LaTeX:

```
\documentclass[PR,CAT]{P3CTeX}
\begin{document}
\section{PR1}
%<*objectius>
\subsection{Objectius}
Text that can also appear in a future memory.
%</objectius>
\section{PR-only work}
Not selected by the memory.
\end{document}
```

The markers are comments in the source document: they do not change its normal output. Only the material between the matching `%<*objectius>` and `%</objectius>` markers is selected. The consumer asks for `objectius`. Do not put document-level commands (`\documentclass`, `\begin{document}`, etc.) inside a region.

4.2 2. \pxMEMsource{id}{source-path}

Registers the path to an existing source `.tex` file under a short identifier. It does not import any material immediately. The path is written explicitly by the consumer, relative to its usual LaTeX working directory; no layout, export directory or filesystem search is imposed.

```
\pxMEMsource{PR1}{../PR1/PR1.tex}
\pxMEMsource{PR2}{../PR2/PR2.tex}
```

4.3 3. \pxMEMinclude{id}{region}

Extracts the named region from the registered source and typesets it at the call site. The rest of the PR remains excluded. A registered source may be included more than once or used in different chapters or annexes.

```
\chapter{Introduction}
\pxMEMinclude{PR1}{objectius}
```

The region is read as ordinary LaTeX after extraction: its macros and referenced packages must be available in the memory document, and included headings, counters and labels act in the consumer document. Avoid copying `\label` names that would conflict with other included material.

4.4 Resource paths and intermediate files

When typesetting reused material, the registered source file's directory is prepended to the normal LaTeX input and graphics search paths for the duration of that inclusion. For instance, `\input{sections/data.tex}` or `\includegraphics{img/chart.png}` may still resolve relative to PR1. The consumer supplies the path; `pxMEM` does not discover it.

Extraction automatically creates a disposable `<jobname>.pxmem-N.tex` file in the compilation working directory, inputs it, and leaves its lifecycle to the usual auxiliary-file cleanup. There is no manual export or regeneration step. Source selection uses the existing `fancyvrb` file-reading mechanism rather than a custom LaTeX parser.

5 Complete generic MEM example

The following document has no TFG dependency and can select regions from two separately compilable assessment documents:

```
\documentclass[MEM,CAT,toc]{P3CTeX}
\begin{document}
\pxMEMsource{PR1}{../PR1/PR1.tex}
\pxMEMsource{PAC2}{../PAC2/PAC2.tex}

\chapter{Context}
\pxMEMinclude{PR1}{objectius}

\chapter{Implementation}
\pxMEMinclude{PAC2}{design}

\pxMEMannex{Detailed evidence}{
  \pxMEMinclude{PAC2}{evidence}
}
\end{document}
```

6 Public API summary

`\pxMEMpreliminaries{content}` Optional manual preliminaries for custom documents with automatic TOC disabled.

`\chapter{title}` Numbered report-like chapter in MEM.

`\pxMEMannex{title}{content}` Letter-numbered annex chapter with arbitrary content.

`\pxMEMsource{id}{path}` Register a known source file.

`\pxMEMinclude{id}{region}` Insert a selected tagged region from a registered source.